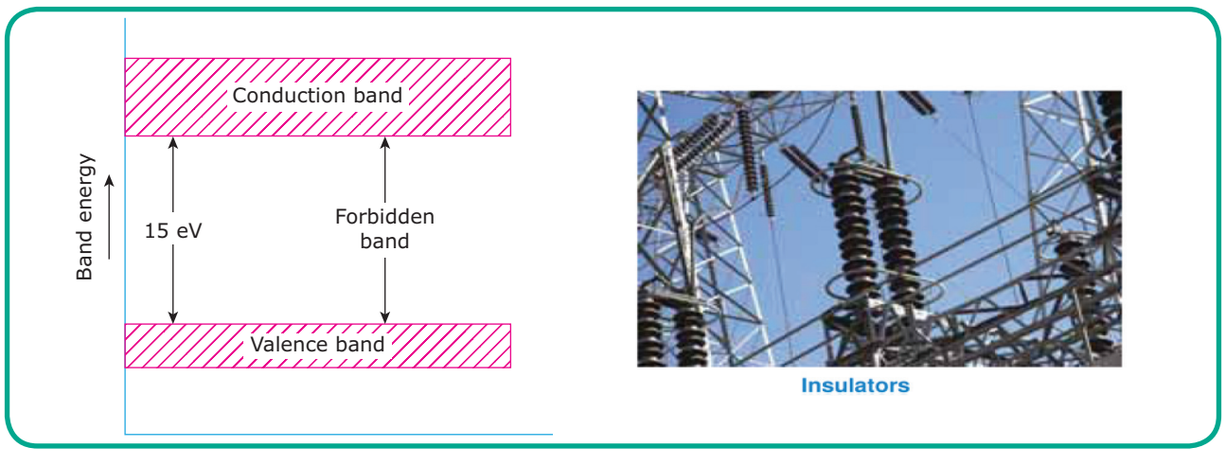


Figure3.15 Insulator and its energy band

(b) Insulators: Figure3.15 shows the forbidden energy gap of the insulators which is very large (15eV), e.g. wood, glass, etc.

c) Semiconductors: Semiconductors (e.g. Germanium, Silicon, Graphene, etc.)are those substance whose electrical property lies in between conductors and insulators. In terms of energy band, the valence band is almost full but the conduction band is empty. Further, the energy gap between valence band and conduction band is very small ($\simeq$, 1eV) as shown in Figure 3.16. Hence, smaller electric field is required to push the electron from the valence band to conduction band.

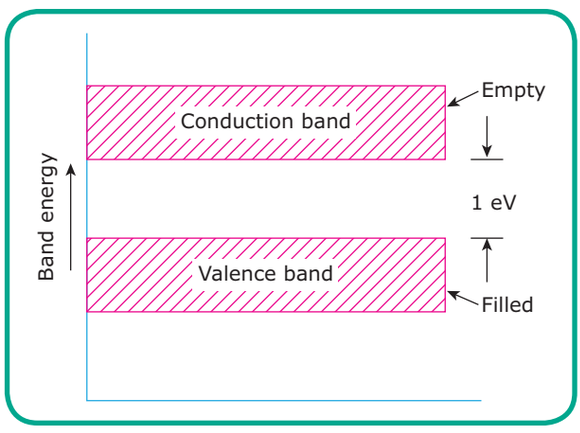


Figure3.16 Energy band of semiconductor

At low temperature, the valence band is completely full and conduction band is completely empty. Therefore at low temperature semiconductor behaves like an insulator. However, even at room temperature some of electrons cross-over to conduction band giving little conductivity to the semiconductor. As temperature increases, more number of electrons cross-over to the conduction band and the conductivity increases. Because of this, the entire characteristics of semiconductors get changed.

3.6 SEMI-CONDUCTOR

In lower standards you might have studied about the characteristics and principle of conductors, insulators. But you may not studied about semiconductors. Based on its character, it has been defined as semiconductor. But now, this semiconductor is the Back Bone of modern electronics. The character of semiconductor lies in between conductor and insulator.

The earlier period of (1950) electronics (communication equipments like Radio, Television and Amplifiers) which was dominated by vacuum tubes, gas filled tubes were replaced by this semiconductors.

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